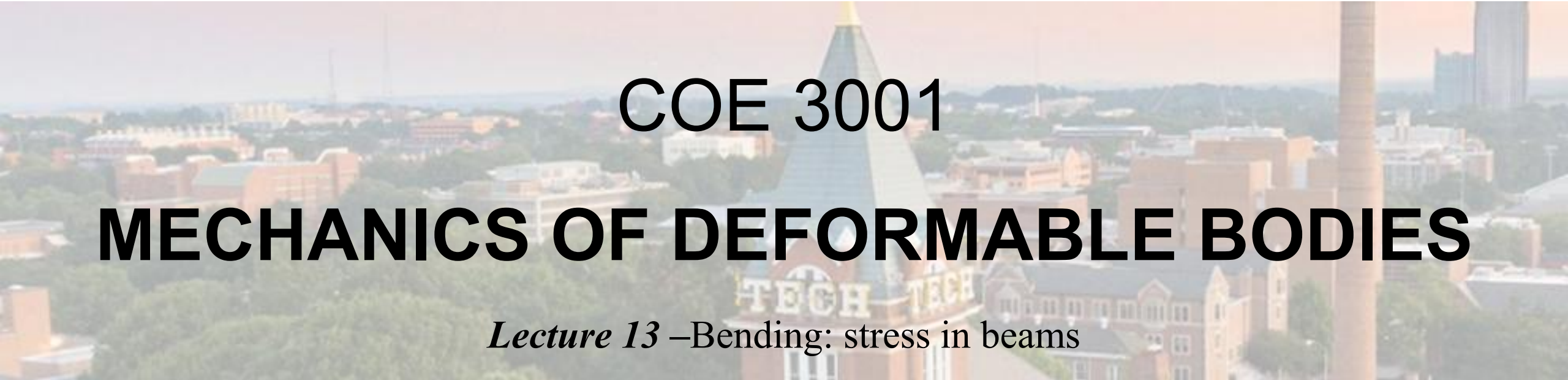




Georgia Tech
College of
Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus, showing various buildings and greenery under a soft, overcast sky.

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 13 –Bending: stress in beams

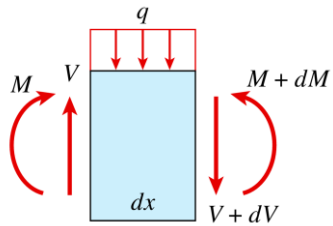
Yazhuo Liu

Georgia Institute of Technology

Mar. 9, 2026

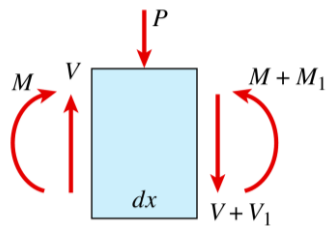
Relationships among $q(x)$, $V(x)$, and $M(x)$

From **left to right** !!!!!

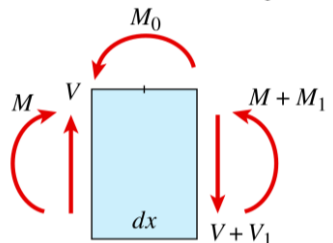


$$\frac{dV}{dx} = -q(x) \quad q(x) \text{ downward}$$

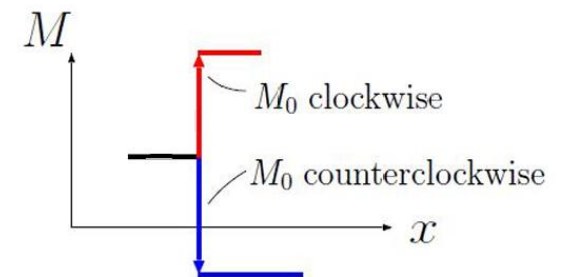
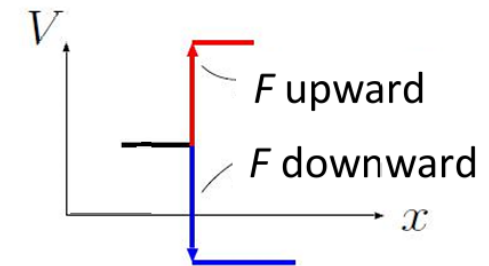
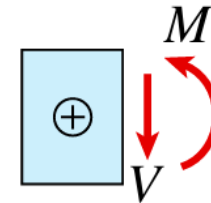
$$\frac{dM}{dx} = V(x)$$



$$\Delta V = -P \quad P \text{ downward}$$



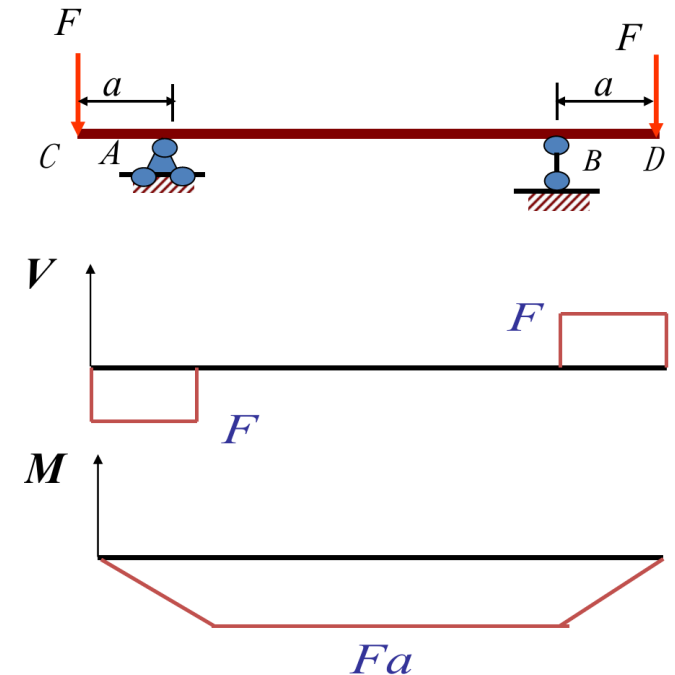
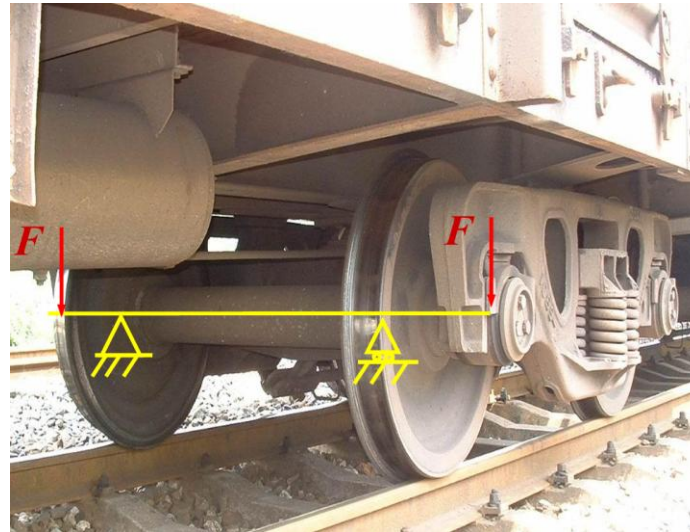
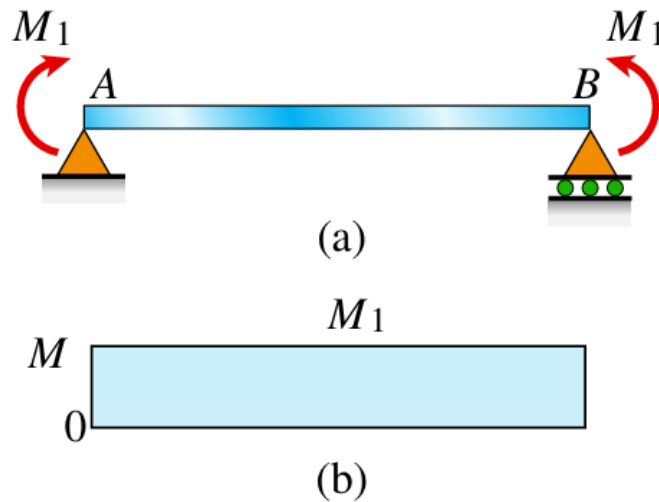
$$\Delta M = -M_0 \quad M_0 \text{ counterclockwise}$$



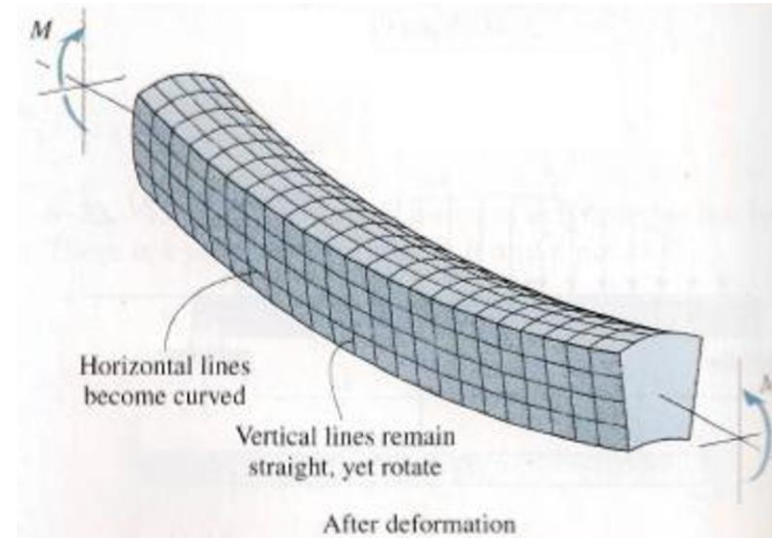
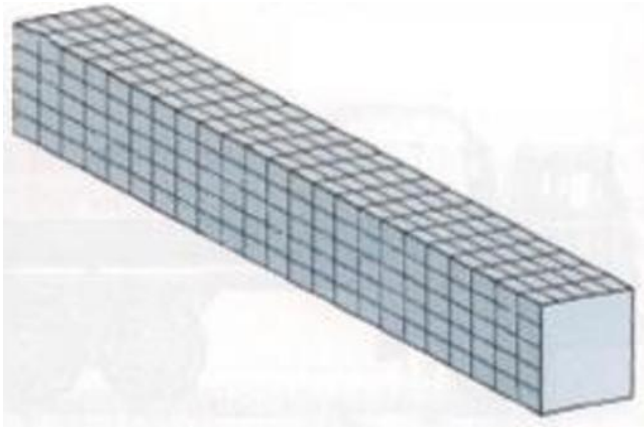
Pure bending

Pure bending refers to the flexure of a beam under a constant bending moment.

- *Nonuniform bending* refers to flexure in the presence of shear forces, which means that the bending moment changes as you move along the axis of the beam.



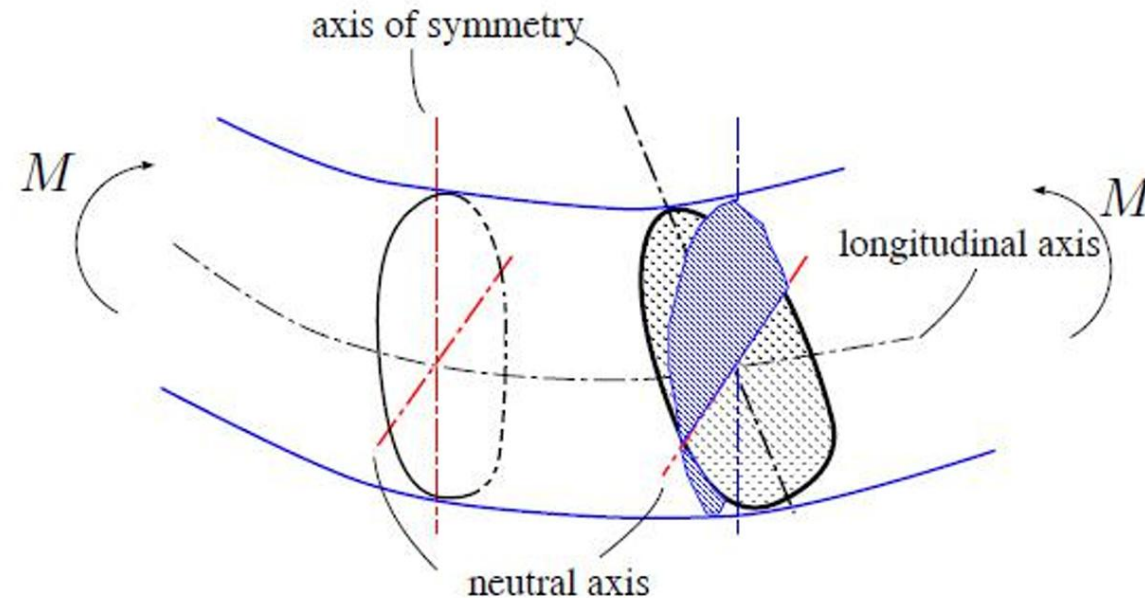
Deformation of a bar due to bending



Observations on deformation:

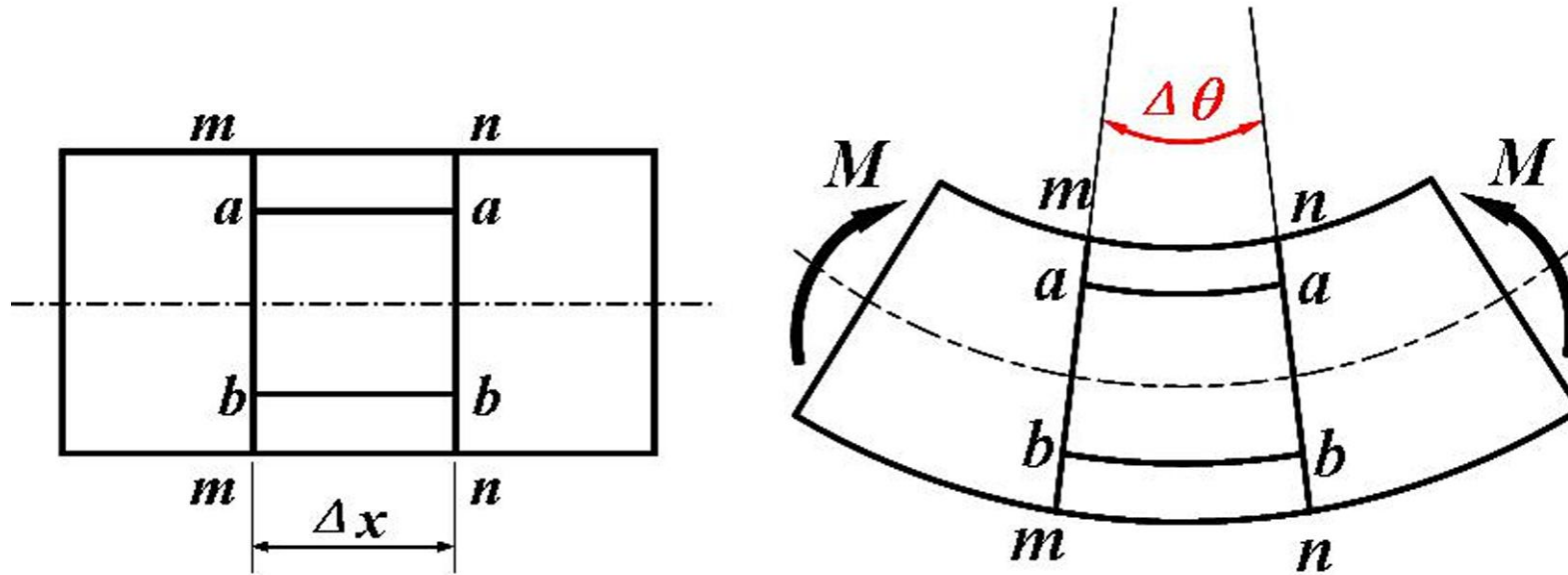
- **longitudinal lines:** straight $\rightarrow$ curved
- **vertical transverse lines:** remain straight with a rotation
- **on cross-section:** stretched and compressed portions

Basic assumptions on pure bending



longitudinal axis undergoes **no change in length**, and stays in the plane of symmetry (bending plane)
all **cross sections** remain **planar** and **perpendicular** to the **longitudinal axis** during deformation
the **in-plane deformation** of the cross section is **neglected**

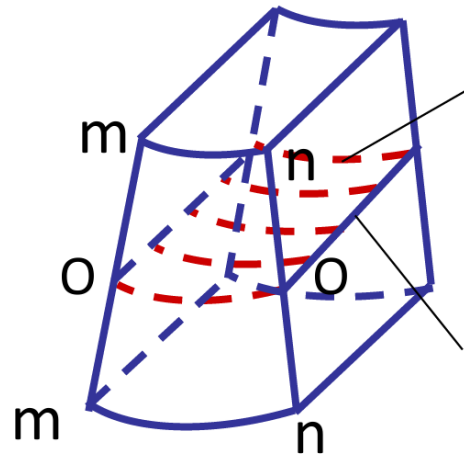
Basic assumptions on pure bending



cross-sectional plane passes through arc center and remains planar

bends uniformly to form a circular arc

Neutral surface and neutral axis



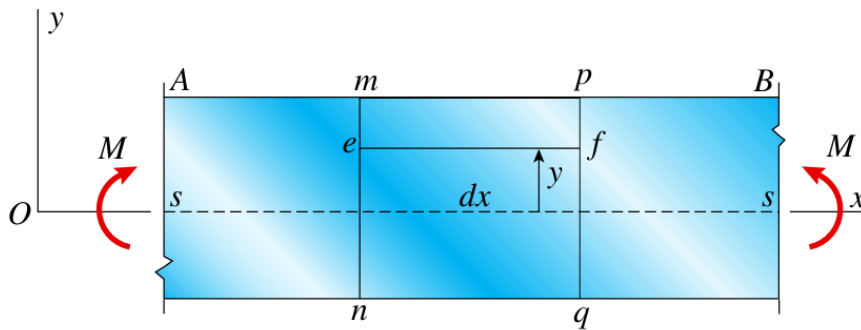
Neutral surface:

The unstretched interface between regions in tension and in compression

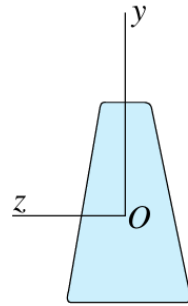
Neutral axis:

Intersection of neutral surface and cross-section

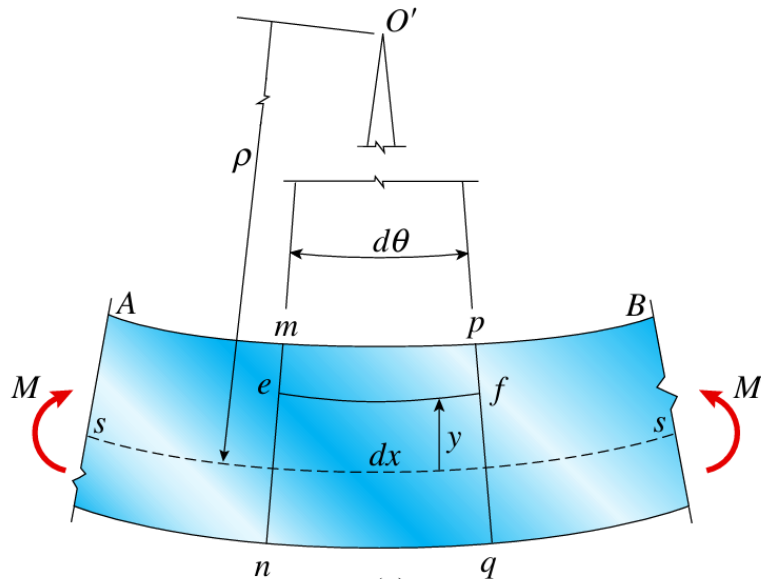
Longitudinal strains in beams



(a)



(b)



(c)

Radius of curvature

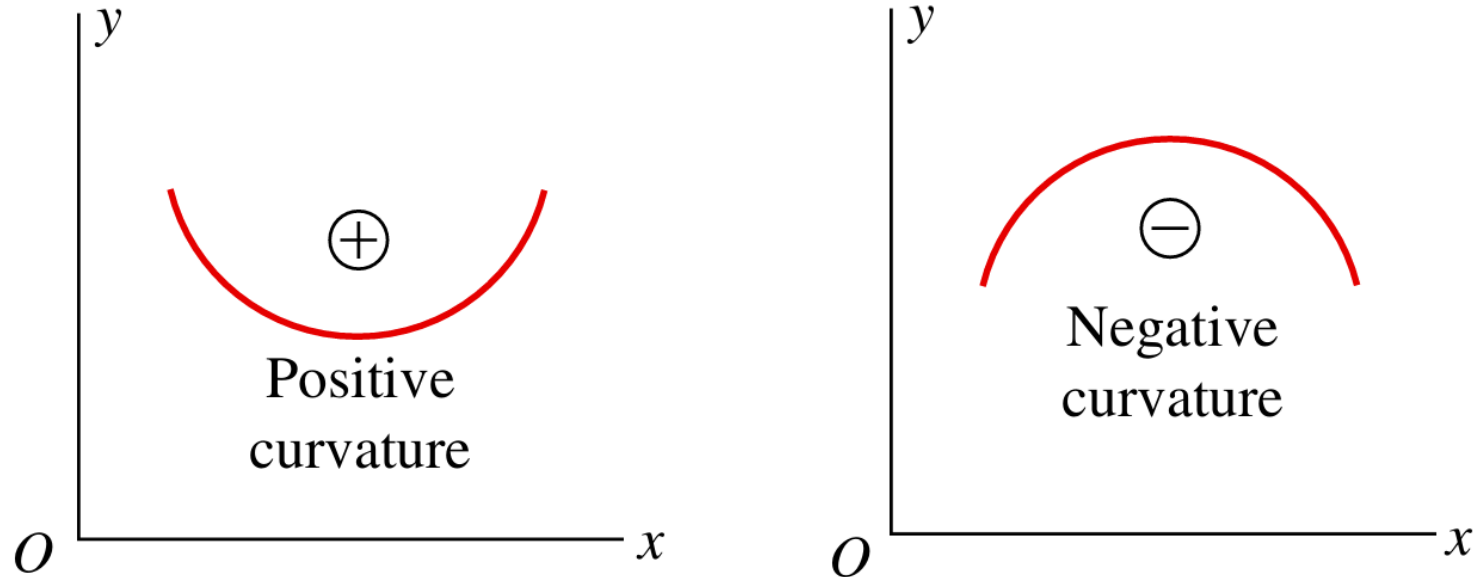
$$\kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$$

$$ef = (\rho - y)d\theta = dx - \frac{y}{\rho}dx$$

$$\epsilon_x = \frac{ef - dx}{dx} = -\frac{y}{\rho} = -\kappa y$$

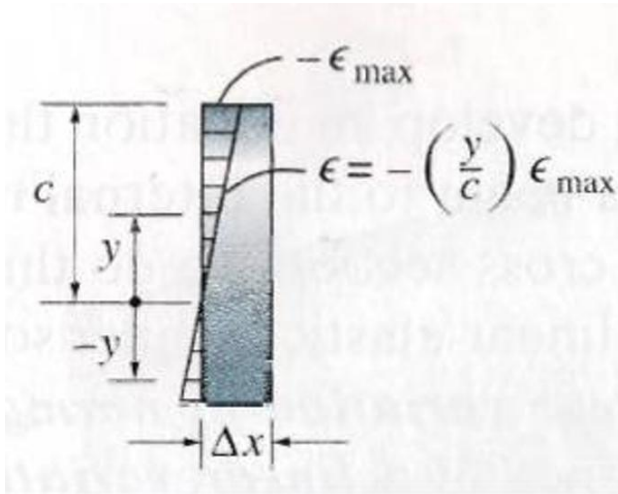
curvature

Sign convention for curvature



The **sign convention for curvature** depends upon the orientation of the coordinate axes. If the x axis is positive to the right and the y axis is positive upward, then the **curvature is positive when the beam is bent concave upward and the center of curvature is above the beam**. Conversely, the curvature is negative when the beam is bent concave downward, and the center of curvature is below the beam.

Linear distribution of longitudinal normal strain

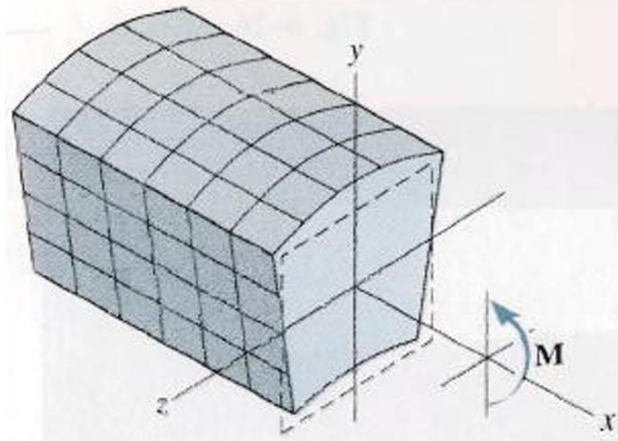


$$\epsilon = -\frac{y}{\rho}$$

$$\epsilon_{\max} = \frac{c}{\rho}$$

$$\frac{\epsilon}{\epsilon_{\max}} = \frac{-y / \rho}{c / \rho} = -\frac{y}{c}$$

$$\epsilon = -\frac{y}{c} \epsilon_{\max}$$



1. We still don't know the location of neutral axis yet.
2. We don't know the value of $\epsilon_{\max}$ yet.

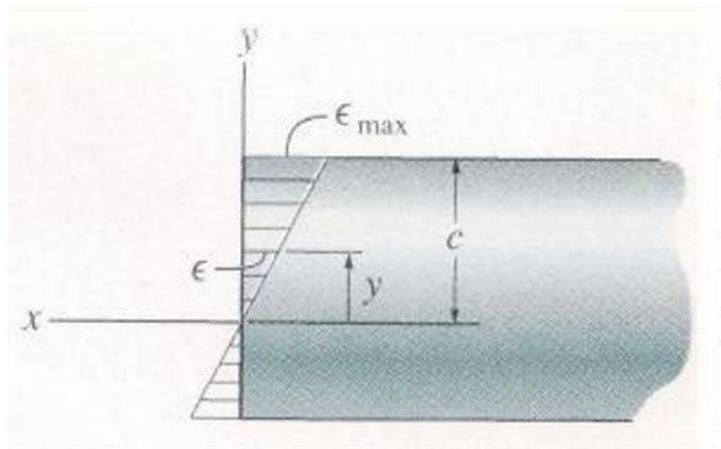
Bending normal stress in beams

For **linear elastic** materials, by Hooke's law

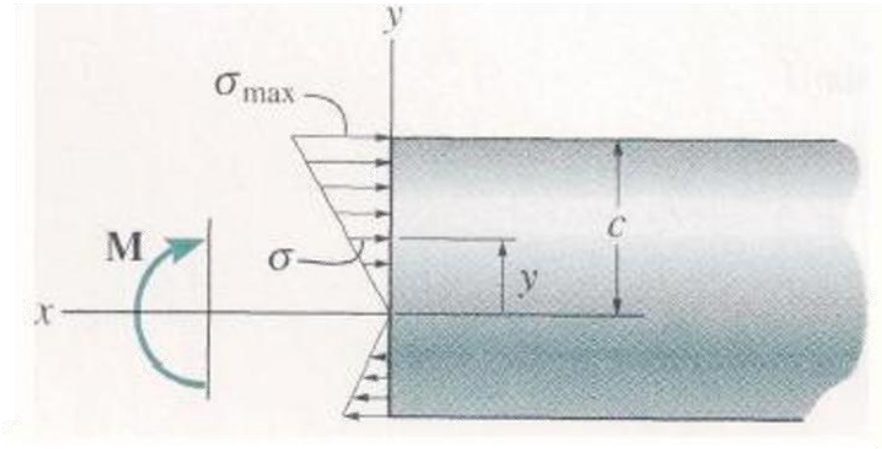
$$\sigma = E\epsilon = E\left(-\frac{y}{\rho}\right) = -E\kappa y$$

Bending stress is again linear in y .

1. We still don't know the location of neutral axis yet.
2. We don't know the value of $\epsilon_{\max}$ yet.

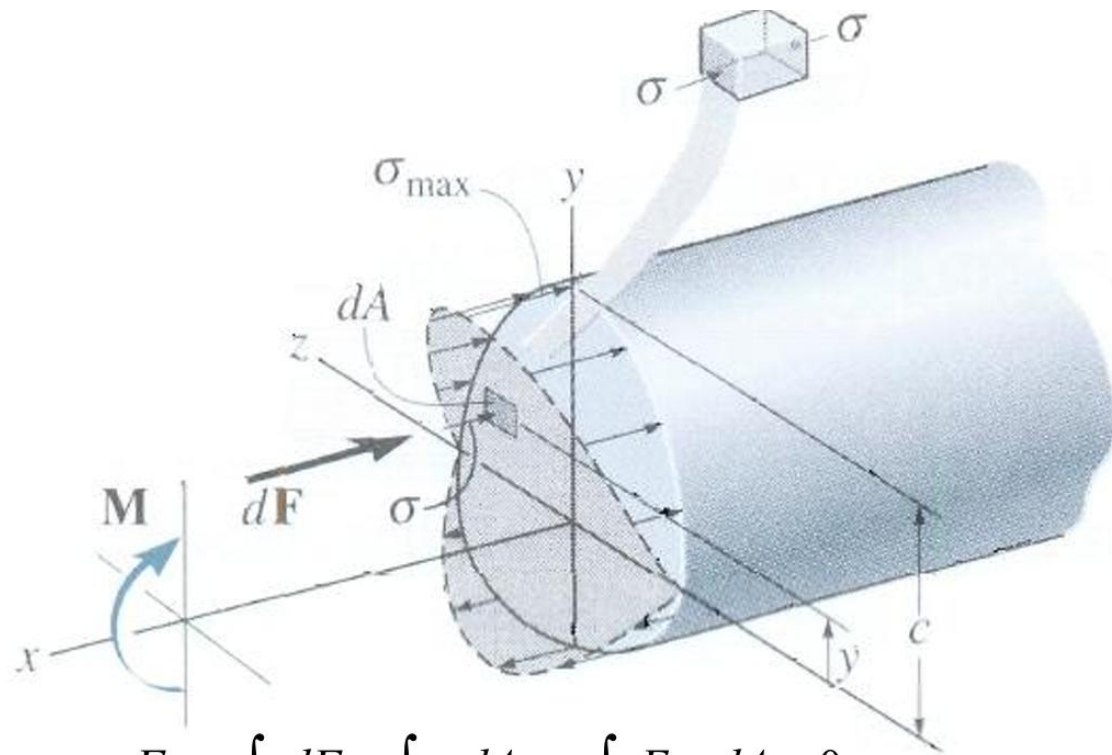


linear bending strain



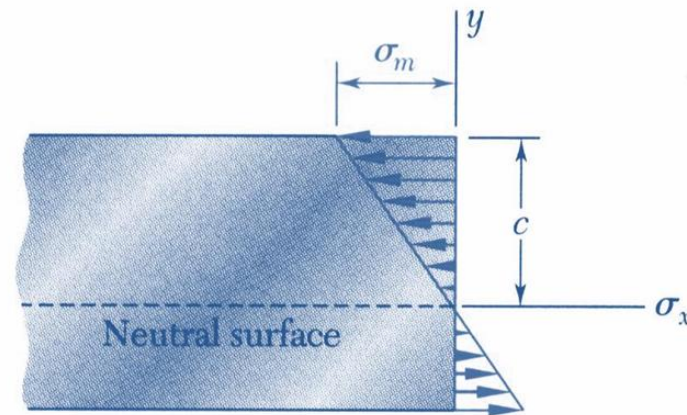
linear bending stress

Location of neutral axis



$$F_N = \int_A dF = \int_A \sigma dA = -\int_A Eky dA = 0$$

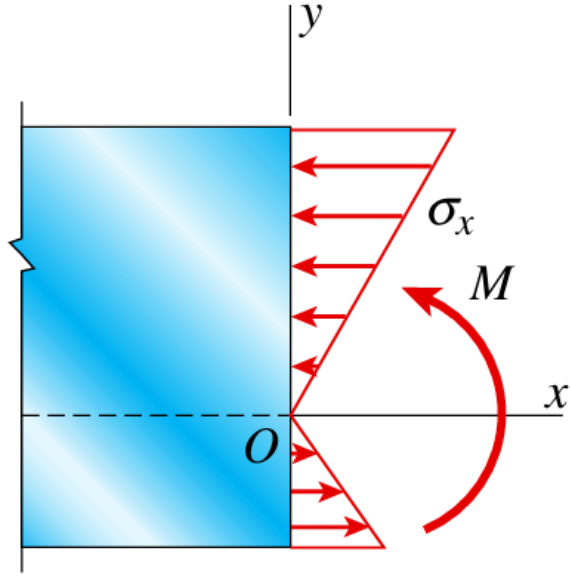
$$\int_A y dA = 0$$



The neutral axis

- Passing through the centroid
- Horizontal centroidal axis

Moment-curvature equation



$$\begin{aligned} M &= -\int_A \sigma_x y dA \\ &= \int_A \kappa E y^2 dA = \kappa E \int_A y^2 dA \end{aligned}$$

$$M = \kappa EI \quad \Rightarrow \quad \frac{M}{EI} = \kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$$

$$I = \int_A y^2 dA$$

Moment-curvature equation

$$dM = -\sigma y dA$$

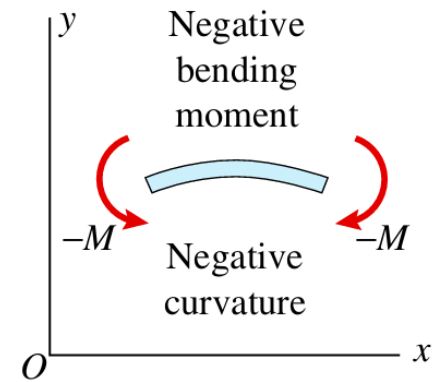
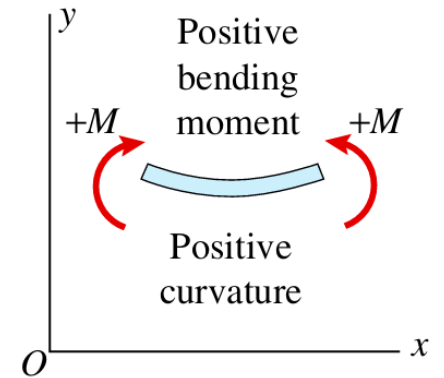
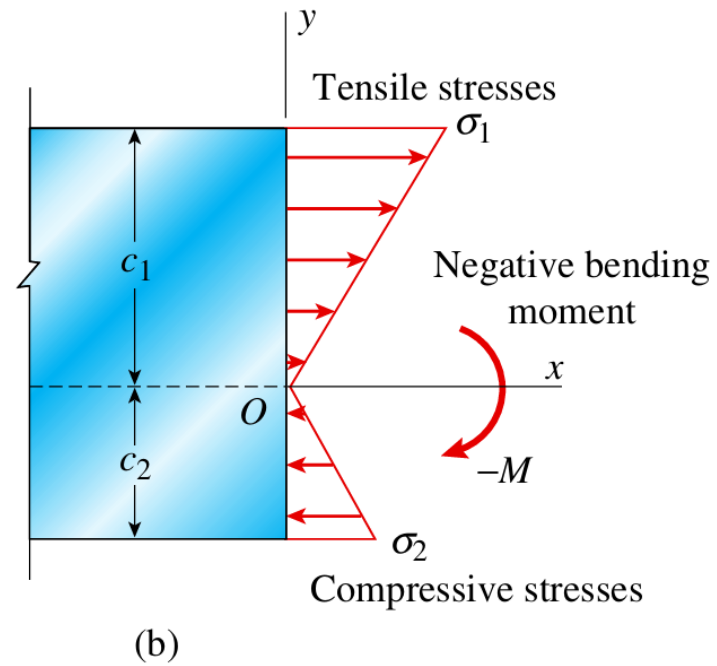
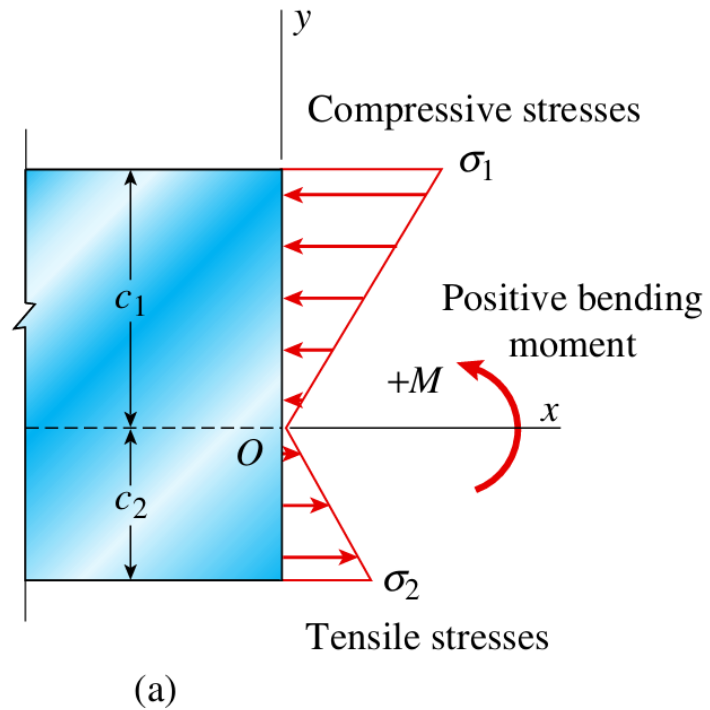
$$y > 0 \quad dM = (-\sigma) y dA$$

$$y < 0 \quad dM = (-y) \sigma dA$$

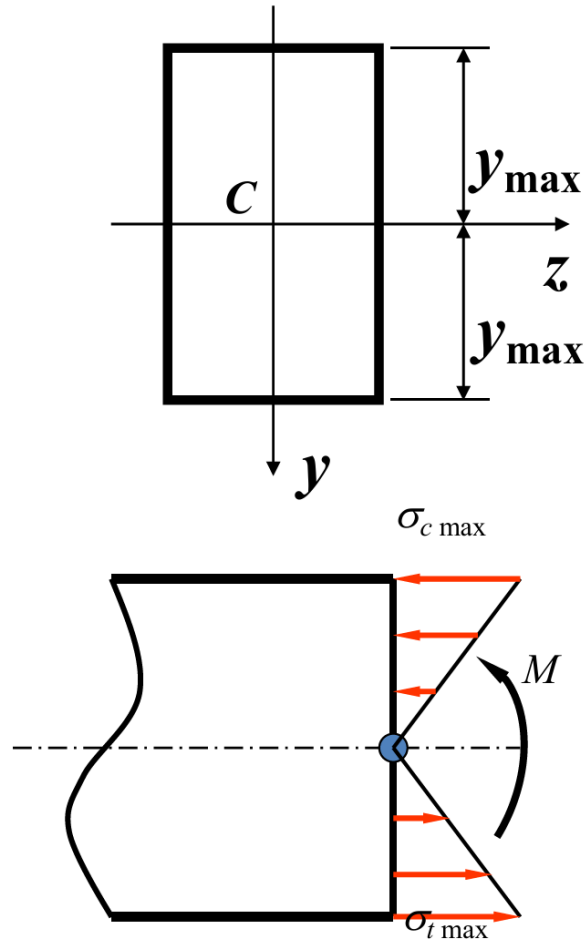
$$\sigma_x = -E \kappa y = -E y \frac{M}{EI} = -\frac{M y}{I}$$

Flexure formula

$$\sigma_x = -\frac{My}{I}$$



Maximum stress analysis



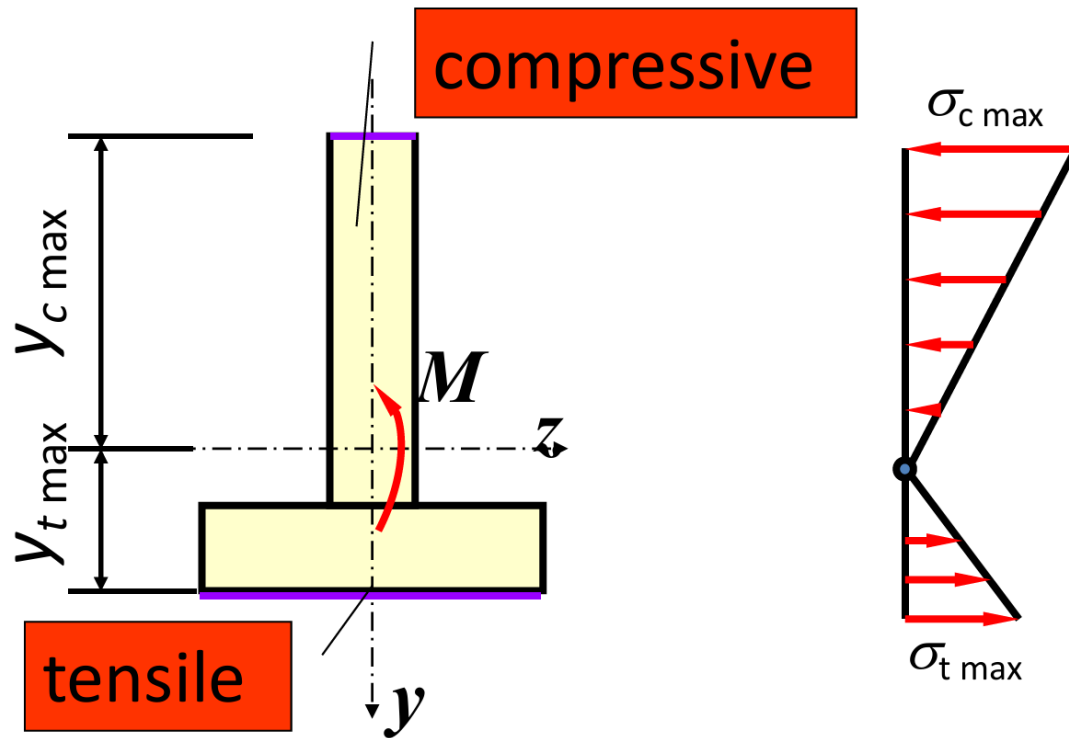
A. neutral axis = symmetry axis

$$\sigma_{\max} = \frac{M_{\max} y_{\max}}{I}$$

$$\sigma_{c \max} = \sigma_{t \max} = \sigma_{\max}$$

Maximum stress analysis

B. neutral axis $\neq$ symmetry axis

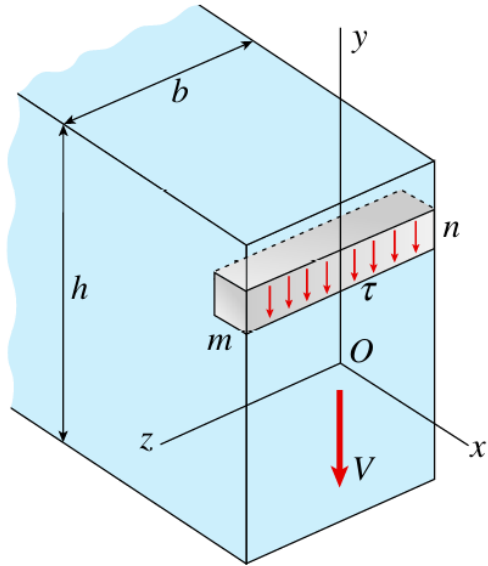


$$\sigma = \frac{My}{I}$$

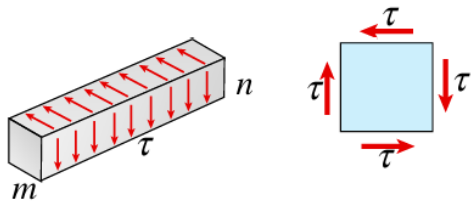
$$\sigma_{t \max} = \frac{My_{t \max}}{I}$$

$$\sigma_{c \max} = \frac{My_{c \max}}{I}$$

Bending shear stress in beams

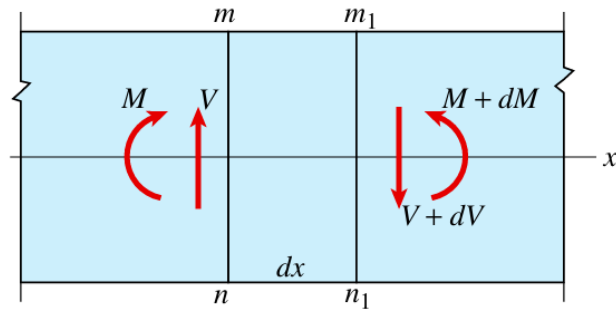


(a)

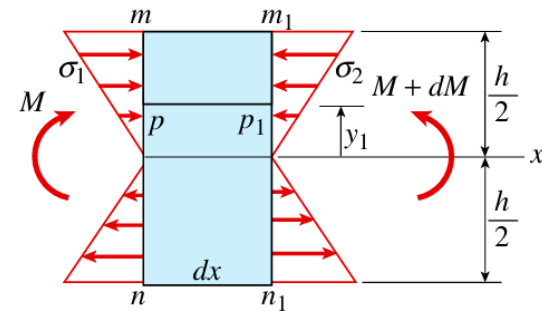


(b)

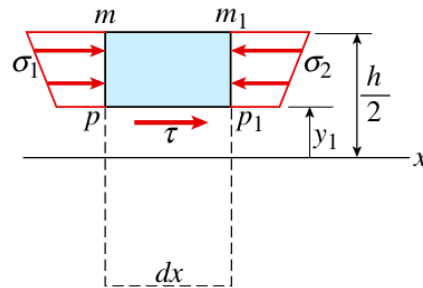
(c)



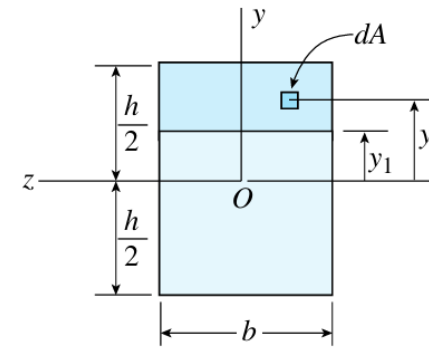
Side view of beam
(a)



Side view of element
(b)

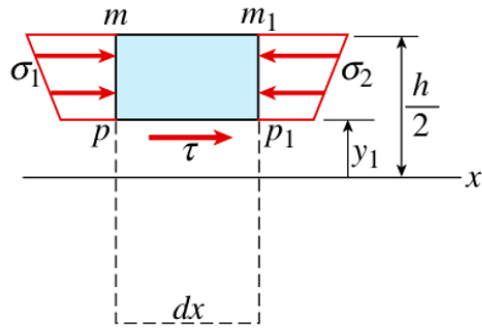


Side view of subelement
(c)



Cross section of beam at subelement
(d)

Bending shear stress in beams



$$F_1 = \int \sigma_1 dA = \int \frac{My}{I} dA$$

$$F_2 = \int \sigma_2 dA = \int \frac{(M + dM)y}{I} dA$$

$$F_3 = F_2 - F_1 = \int \frac{(dM)y}{I} dA = \frac{dM}{I} \int y dA$$

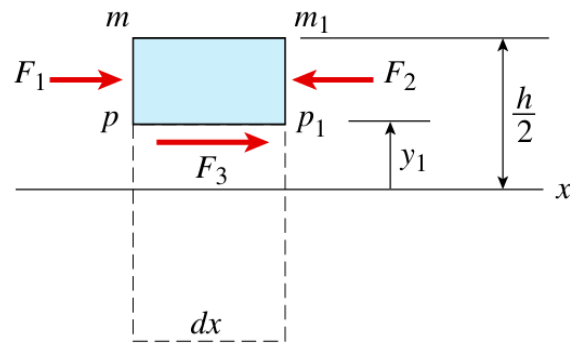
Also

$$F_3 = \tau b dx$$

$$\Rightarrow \tau = \frac{dM}{dx} \frac{1}{Ib} \int y dA = \frac{V}{Ib} \int y dA$$

Denote

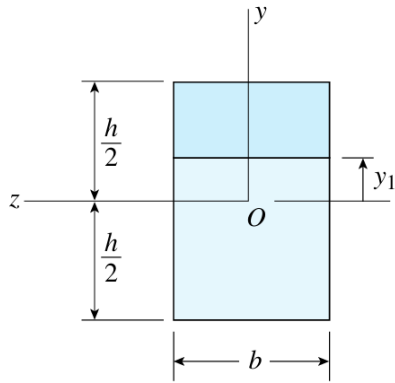
$$Q = \int y dA \Rightarrow \tau = \frac{VQ}{Ib}$$



$$\sigma_1 = -\frac{My}{I}$$

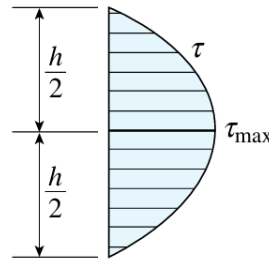
$$\sigma_2 = -\frac{(M + dM)y}{I}$$

Rectangular Beam



$$Q = \int y dA = \int_{y_1}^{\frac{h}{2}} y b dy = \frac{b}{2} y^2 \Big|_{y_1}^{\frac{h}{2}} = \frac{b}{2} \left(\frac{h^2}{4} - y_1^2 \right)$$

$$\Rightarrow \tau = \frac{VQ}{Ib} = \frac{V}{2I} \left(\frac{h^2}{4} - y_1^2 \right)$$



$$\tau_{\max} = \frac{Vh^2}{8I} = \frac{3V}{2A}$$

Warping of the cross sections of a beam due to shear strains

